

Abhijith Thakur

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Summary

Data Engineer with 3+ years of experience in ETL data pipeline development and machine learning integration across healthcare and IT consulting industries. Expertise in both AWS and Azure cloud platforms, migrated legacy ETL workflows to AWS using Glue and Lambda, reducing data latency by 30% and improving analytics turnaround for healthcare operations. Implemented ML models for treatment predictions using TensorFlow on Sage Maker, increasing early intervention rates by 18%. Proficient in Azure cloud services including Azure Data Factory, Azure SQL Database, Azure Data Lake Storage, and Azure Blob Storage. Expertise in Python, SQL, PowerBI, Pyspark, and Tableau, holding certifications including Azure Data Engineer Associate and AWS Data Engineer Associate and authorized to work in the US.

Skills

- **Programming Language:** Python (Pandas, NumPy), Java, C, Spark
- **Data Engineering & Tools:** Azure Data Factory, Azure Data Lake, PySpark, Apache Airflow, Kinesis, Apache Kafka, Hadoop, Databricks, Snowflake Initio, ETL/ELT, Alteryx
- **Machine Learning & Analytics:** TensorFlow, PyTorch, Model Deployment (Azure ML, Sage Maker), Predictive Analytics, A/B Testing, Anomaly Detection
- **Database Management:** MySQL, PostgreSQL, Spark SQL, NO SQL
- **Cloud Platforms:** AWS (S3, EMR, Lambda, Athena, Redshift), Azure (Data Factory, Synapse, Data Lake Gen 2)
- **Data Visualization:** Tableau, Power BI, KPI Dashboard Development, Excel
- **Version Control:** Git, GitHub
- **Project Management:** Agile, Scrum, JIRA, SDLC, Confluence

Work Experience

Cigna Healthcare Data Engineer

Plano, TEXAS
Sep 2024 – Present

- Established a repeatable, automated deployment process for AWS Glue jobs, resulting in zero failed deployments and improving data pipeline reliability for healthcare operations teams
- Built scalable AWS-based data pipelines integrating S3, Redshift, and Athena to support ingestion and processing of over 5TB/day of clinical and claims data
- Constructed data integrity checks using Python (Pandas, NumPy), improving the consistency of 10+ high-priority data pipelines, thus improving the reliability of healthcare operations reports by 40%
- Orchestrated nightly ETL jobs across distributed healthcare datasets via Apache Airflow, cutting data processing time by 4 hours and improving report generation speed by 35%
- Developed patient analytics dashboards in Tableau and Power BI by integrating insights from AWS Athena queries, enhancing care strategies
- Built machine learning models with TensorFlow and SageMaker, enabling a 20-bed increase in hospital capacity through better resource allocation based on patient needs
- Streamlined repetitive data tasks with Python scripts, utilizing libraries such as Pandas and NumPy for data cleaning and transformation, increasing efficiency and reducing manual effort by 40%

TATA Consultancy Services - TCS Data Engineer

Hyderabad, INDIA
Jul 2021 – Dec 2022

- Mastered Ab Initio, Python, and SQL to build complex data models and ETL processes, increasing personal data processing speed by 20% within the first quarter
- Created cloud-native ETL solutions on Azure and AWS for enterprise clients, integrating Blob Storage, Azure Functions, AWS Glue, and Redshift
- Redesigned and maintained data lakes using AWS S3 and implemented Glue Crawlers for schema inference, reducing data cataloging time by 50%
- Launched weekly data integrity checks on 500+ files using Great Expectations, which supported the migration of data to a new S3 data lake, preventing downstream errors
- Integrated ML frameworks into data pipelines for fraud detection and sales forecasting, leveraging Azure Databricks and Spark for real-time analytics
- Enhanced data pipeline robustness by composing comprehensive unit tests with PyTest, resulting in a 60% reduction in data corruption errors and faster issue resolution
- Collaborated with cross-functional teams in an agile environment, contributing to a 15% improvement in project delivery timelines

- Pioneered ETL solutions using Python and SQL to transform data from CRM/ERP platforms, reaching 95% data consistency; this enabled the processing of 500,000+ records with zero data loss daily
- Examined CRM and ERP data using SQL, identifying process flaws diminishing data quality by 40%; presented recommendations to rectify the three major sources of reporting inaccuracies
- Delivered ad-hoc SQL queries and dashboard monitoring for marketing and finance, directly informing strategic pivots that boosted campaign ROI by 15% and cut costs by 8%
- Engineered marketing analytics projects targeting customer behavior, improving customer lifetime value (CLTV) through personalized campaign messaging based on A/B test insights
- Developed initial prototypes for anomaly detection systems using Pandas, NumPy, and Scikit-learn, providing foundational models for further development
- Maintained comprehensive documentation for five critical data integration workflows using Confluence and Lucid chart, ensuring seamless knowledge transfer and resulting in zero data loss incidents

Certifications

- DP 203: AZURE DATA ENGINEER – ASSOCIATE
- AWS DATA ENGINEER – ASSOCIATE
- DATA ANALYST – ONE ROADMAP
- DATABASE PROGRAMMING WITH SQL – ORACLE ACADEMY
- DATABRICKS FUNDAMENTALS ACCREDITATION

Research Projects

Density Based Smart Traffic Control System Using CNN

- Constructed an intelligent traffic management system applying convolutional neural networks to optimize signal timings and diminishing traffic accidents at monitored intersections to zero in 6 months
- Executed traffic density classification using CNNs on intersection video feeds, leading to findings to fix the three biggest causes of crashes, improving overall road safety
- Integrated a predictive algorithm into the traffic control system, achieving a measurable improvement in throughput, with a documented increase of 800 vehicles processed per hour during rush hour

Education

- **Lamar University** Beaumont, Texas
Master of Science in Computer Science
- **Jawaharlal Nehru Technological University** Hyderabad, India
Bachelor of Science in Computer Science